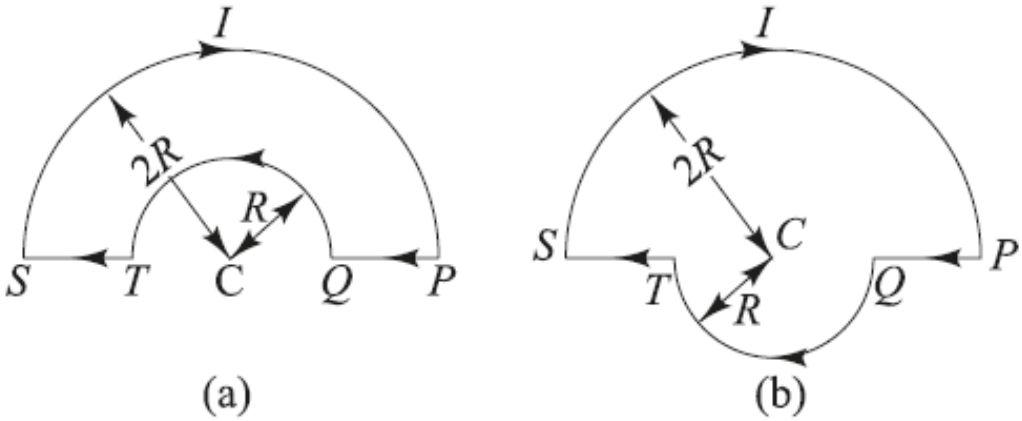
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A wire loop $PQTS$ is formed by joining two semicircular wires of radii R and $2R$ in two different ways as shown in Fig. (a) and (b). They carry the same current I. The ratio of magnitudes of magnetic fields at centre C in case (a) to that in case (b) is



(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{1}{4}$

Q.2

A long straight wire of radius a carries a steady current I. The current is uniformly distributed over its cross-section. The ratio of the magnetic fields B and B', at radial distances $\frac{a}{2}$ and 2 a respectively, from the axis of the wire is:

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 1

(D) 4

Q.3

An electron moving in a circular orbit of radius makes n rotations per second. The magnetic field produced at the centre has magnitude:

(A) Zero

(B) $\frac{\mu_0 n^2 e}{r}$

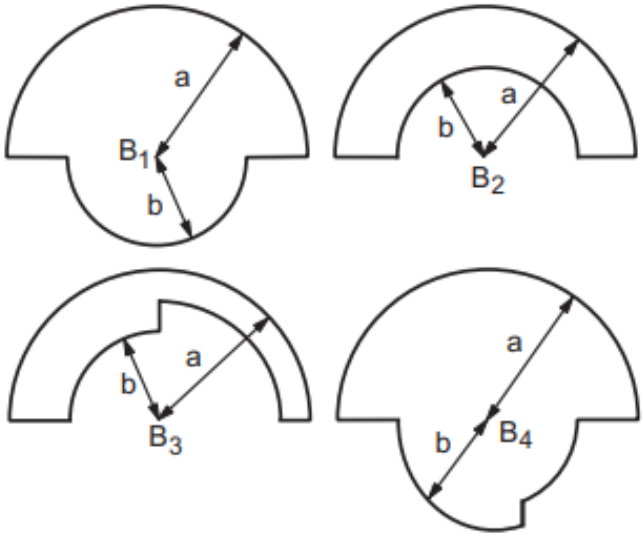
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(C) $\frac{\mu_0 \text{ ne}}{2r}$

(D) $\frac{\mu_0 ne}{2\pi r}$

Q.4

In the loops shown in figure. all curved sections are either semi-circles or quarter circles. All the loops carry same current. The magnetic fields at the centres have magnitudes $\mathrm{B}_1,\ \mathrm{B}_2,\ \mathrm{B}_3$ and B_4 Then, which is correct?



(A) $\mathrm{B}_1 > \mathrm{B}_2 > \mathrm{B}_3 > \mathrm{B}_4$

(B) $\mathrm{B}_3 > \mathrm{B}_4 > \mathrm{B}_1 > \mathrm{B}_1$

(C) $\mathrm{B}_4 > \mathrm{B}_1 > \mathrm{B}_2 > \mathrm{B}_3$

(D) $\mathrm{B}_1 > \mathrm{B}_4 > \mathrm{B}_3 > \mathrm{B}_2$

Q.5

A current loop consists of two identical semicircular parts each of radius R, one lying in the $x-y$ plane and the other in $x-z$ plane. If the current in the loop is i., the resultant magnetic field due to the two semicircular parts at their common centre is

(A) $\frac{\mu_0 i}{\sqrt{2}R}$

(B) $\frac{\mu_0 i}{2\sqrt{2}R}$

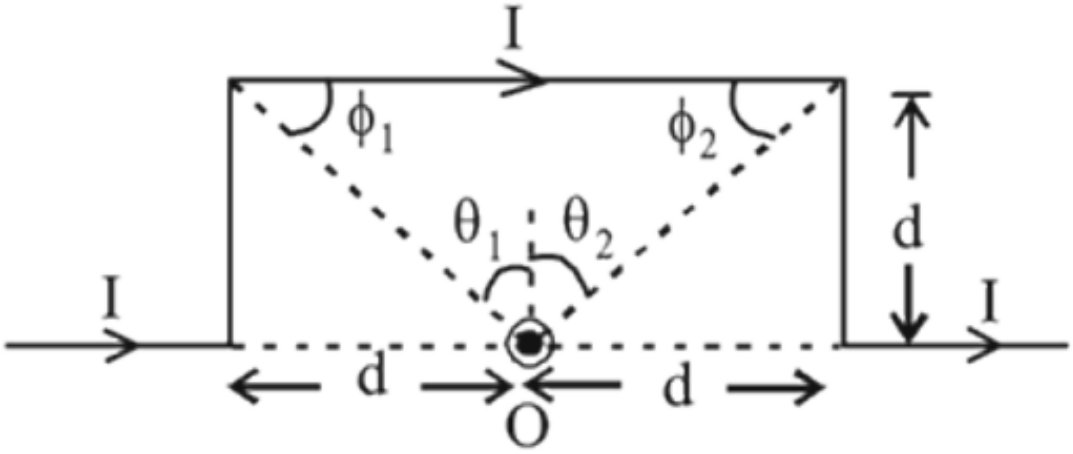
(C) $\frac{\mu_0 i}{2R}$

(D) $\frac{\mu_0 i}{4R}$

Q.6

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The magnetic field at O due to current in the infinite wire forming a loop as shown in Fig. is



(A) $\frac{\mu_0 \mathrm{I}}{2\pi \mathrm{d}}(\cos\phi_1 + \cos\phi_2)$

(B) $\frac{\mu_0 \mathrm{I2I}}{4\pi \mathrm{d}}(\tan\theta_1 + \tan\theta_2)$

(C) $\frac{\mu_0 \mathrm{I}}{4\pi \mathrm{d}}(\sin\phi_1 + \sin\phi_2)$

(D) $\frac{\mu_0 \mathrm{I}}{4\pi \mathrm{d}}(\cos\theta_1 + \sin\theta_2)$

Q.7

A current I flows through a thin wire shaped as regular polygon of n sides which can be inscribed in a circle of radius R. The magnetic field induction at the center of polygon due to one side of the polygon is

(A) $\frac{\mu_0 \mathrm{I}}{\pi \mathrm{R}}\left(\tan\frac{\pi}{\mathrm{n}}\right)$

(B) $\frac{\mu_0 \mathrm{I}}{4\pi R}\left(\tan\frac{\pi}{\mathrm{n}}\right)$

(C) $\frac{\mu_0 \mathrm{I}}{2\pi \mathrm{R}}\left(\tan\frac{\pi}{\mathrm{n}}\right)$

(D) $\frac{\mu_0 \mathrm{I}}{2\pi \mathrm{R}}\left(\cos\frac{\pi}{\mathrm{n}}\right)$

Q.8

A thin rod is bent in the shape of a small circle of radius r. If the charge per unit length of the rod is σ, and if the circle is rotated about its axis at a rate of n rotations per second, the magnetic induction at a point on the axis at a large distance y from the centre

(A) $\mu_0 \pi r^3 n \frac{\sigma}{y^3}$

(B) $2\mu_0 \pi r^3 n \frac{\sigma}{y^3}$

(C) $\left(\frac{\mu_0}{4\pi}\right) r^3 n \frac{\sigma}{y^3}$

(D) $\left(\frac{\mu_0}{2\pi}\right) r^3 n \frac{\sigma}{y^3}$

Q.9

Two identical long conducting wires AOB and COD are placed at right angle to each other, with one above other such that ' O ' is their common point for the two. The wires carry I_1 and I_2 currents respectively. Point '

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P 'is lying at distance 'd' from ' O ' along a direction perpendicular to the plane containing the wires. The magnetic field at the point 'P' will be :

(A) $\frac{\mu_0}{2\pi \mathrm{d}}\left(\frac{\mathrm{I}_1}{\mathrm{I}_2}\right)$

(B) $\frac{\mu_0}{2\pi \mathrm{d}}\left(\mathrm{I}_1+\mathrm{I}_2\right)$

(C) $\frac{\mu_0}{2\pi d}\left(\mathrm{I}_1^2-\mathrm{I}_2^2\right)$

(D) $\frac{\mu_0}{2\pi \mathrm{d}}\left(\mathrm{I}_1^2+\mathrm{I}_2^2\right)^{1/2}$

Q.10

Two similar coils of radius R are lying concentrically with their planes at right angles to each other. The currents flowing in them are I and $2I$, respectively. The resultant magnetic field induction at the centre will be:

(A) $\frac{\sqrt{5}\mu_0 I}{2R}$

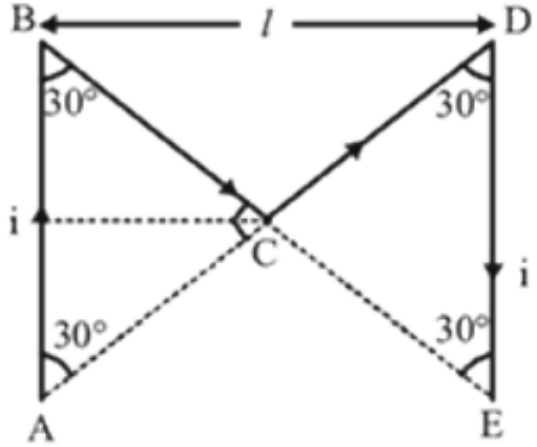
(B) $\frac{3\mu_0 I}{2R}$

(C) $\frac{\mu_0 I}{2R}$

(D) $\frac{\mu_0 I}{R}$

Q.11

The magnetic field at point ' C' due to current flowing in 'M' shape figure is



(A) $\frac{\mu_0}{2\pi} \cdot \frac{\sqrt{3}\mathrm{i}}{\ell}$

(B) $\frac{\mu_0}{\pi} \cdot \frac{\mathrm{i}}{\ell}\sqrt{3}$

(C) zero

(D) $\frac{\mu_0}{4\pi} \cdot \frac{\mathrm{i}}{\ell\sqrt{3}}$

Q.12

A current of 5 ampere is passed through a straight, wire of length $6\ \mathrm{cm}$; then the magnetic induction at a point $5\ \mathrm{cm}$ from the both end of the wire is

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(A) 0.25 gauss

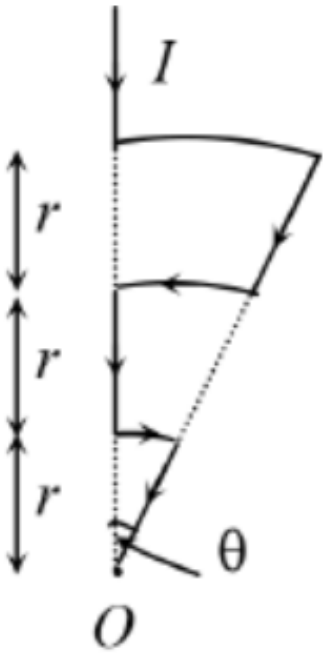
(B) 0.125 gauss

(C) 0.15 gauss

(D) 0.30 gauss

Q.13

Shown in the figure is a conductor carrying a current I The magnetic field intensity at the point O (common centre of all the three arcs) is $(\theta$ in radian)



(A) $\frac{5\mu_0 I\theta}{24\pi r}$

(B) $\frac{\mu_0 I}{24\pi}$

(C) $\frac{11\mu_0 I\theta}{24\pi r}$

(D) zero

Q.14

A circular current carrying loop of radius $\mathbf{R}$, carries a current $\mathbf{i}$. The magnetic field at a point on the axis of coil is $\frac{1}{\sqrt{8}}$ times the value of magnetic field at the centre. Distance of point from centre is

(A) $\frac{R}{\sqrt{2}}$

(B) $\frac{R}{\sqrt{3}}$

(C) $R\sqrt{2}$

(D) R

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Q.15

A long straight wire carrying a current of $30A$ is placed in an extenal uniform magnetic field of magnitude 4×10^{-4} T. The magnetic field is acting parallel to the direction of curent. The magnitude of the resultant magnetic field in tesla at a point 2.0 cm away from the wire is

(A) 10^{-4}

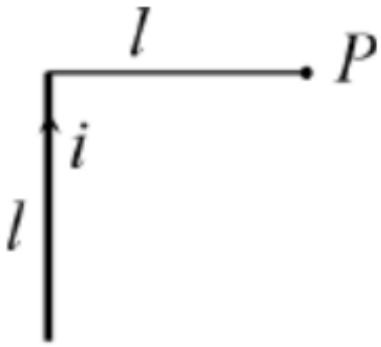
(B) 3×10^{-4}

(C) 5×10^{-4}

(D) 6×10^{4}

Q.16

Figure shows a straight wire of length l carrying a current i. The magnitude of magnetic field produced by the current at point P is



(A) $\frac{\sqrt{2\mu_0 i}}{\pi l}$

(B) $\frac{\mu_0 i}{4\pi l}$

(C) $\frac{\sqrt{2}\mu_0 i}{8\pi l}$

(D) $\frac{\mu_0 i}{2\sqrt{2}\pi l}$

Q.17

Two long parallel wires carry currents i_1 andn i_2 such that $\mathrm{i}_1 > \mathrm{i}_2$. When the currents are in the same direction, the magnetic field at a point midway between the wires is 6×10^{-6} T. If the direction of i_2 is reversed, the field becomes 3×10^{-5} T. The ratio of $\frac{\mathrm{i}_1}{\mathrm{i}_2}$ is

(A) $\frac{1}{2}$

(B) 2

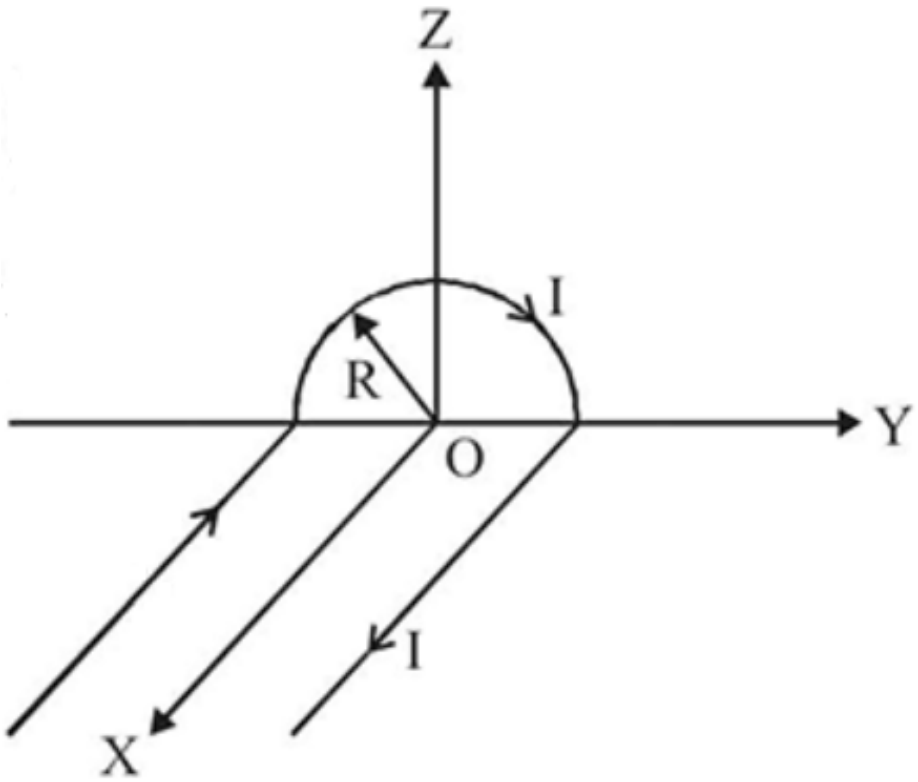
(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

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Q.18

A wire carrying current I has the shape as shown in adjoining figure. Linear parts of the wire are very long and parallel to X-axis while semicircular portion of radius R is lying in $\mathrm{Y}-\mathrm{Z}$ plane. Magnetic field at point O is :



(A) $\overrightarrow{\mathrm{B}}=-\frac{\mu_0}{4\pi}\frac{\mathrm{I}}{\mathrm{R}}(\mu\hat{\mathrm{i}}\times 2\hat{\mathrm{k}})$

(B) $\overrightarrow{\mathrm{B}}=-\frac{\mu_0}{4\pi}\frac{\mathrm{I}}{\mathrm{R}}(\hat{\pi}\hat{\mathrm{i}}+2\hat{\mathrm{k}})$

(C) $\overrightarrow{\mathrm{B}}=\frac{\mu_0}{4\pi}\frac{\mathrm{I}}{\mathrm{R}}(\lambda\hat{\mathrm{i}}-2\hat{\mathrm{k}})$

(D) $\overrightarrow{\mathrm{B}}=\frac{\mu_0}{4\pi}\frac{\mathrm{I}}{\mathrm{R}}(\pi\hat{\mathrm{i}}+2\hat{\mathrm{k}})$

Q.19

A closely wound solenoid of 2000 turns and area of cross section $1.5\times10^{-4}\ \mathrm{m}^2$ carries a current of $2.0A$. It suspended through its centre and perpendicular to its length, allowing it to turn in a horizontal plane in a uniform magnetic field 5×10^{-2} tesla making an angle of 30° with the axis of the solenoid. The torque on the solenoid will be:

(A) $3\times10^{-2}\ \mathrm{N}-\mathrm{m}$

(B) $3\times10^{-3}\ \mathrm{N}-\mathrm{m}$

(C) $1.5\times10^{-3}\ \mathrm{N}-\mathrm{m}$

(D) $1.5\times10^{-2}\ \mathrm{N}-\mathrm{m}$

Q.20

For a long cylindrical wire of radius $a=12$ mm, we know that its current density $\vec{J}$ is along the direction of

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the central axis and its magnitude varies linearly with radial distance r from the axis according to the relation $J = \frac{J_0 r}{a}$, where $J_0 = \frac{10^5}{4\pi}$ A / m^2. Calculate the magnitude of the magnetic induction at a position $r = \frac{a}{2}$(in μT).

(A) 10

(B) 20

(C) 30

(D) 40

Answers & Solutions

Q.1 Answer:

$\frac{1}{3}$

Solution:

The magnetic field due to straight segments PQ and TS in both cases in zero.

Case (a)

The magnetic field at C due to semi-circular loop of radius R is $B = \frac{\mu_0 I}{4R}$ directed out of the page, since the current is clockwise. The magnetic field at C due to semi-circular loop of radius $2R$ is $B' = \frac{\mu_0 I}{4(2R)} = \frac{\mu_0 I}{8R}$ directed into the page since the current now is anticlockwise. Since $B > B'$, the net magnetic field at C is

$$B_1 = B - B' = \frac{\mu_0 I}{4}\left(\frac{1}{R} - \frac{1}{2R}\right) = \frac{\mu_0 I}{8R}$$

directed out of the page.

Case (b) In this case, the current in both loops is clockwise. Therefore, the magnetic field due to each semicircular loop is directed into the page and its magnitude is

$$B_1 = B + B' = \frac{\mu_0 I}{4}\left(\frac{1}{R} + \frac{1}{2R}\right) = \frac{3\mu_0 I}{8R}$$

directed into the page. Therefor, $\frac{B_1}{B_2} = \frac{1}{3}$, which is choice (c).

Q.2 Answer:

1

Solution:

Consider two amperian loops of radius a/ 2 and 2 a as shown in the diagram. Applying ampere's circuital law for these loops, we get

$$\oint \mathrm{B} \cdot \mathrm{dL} = \mu_0 \mathrm{I}_{\text{enclosed}}$$

For the smaller loop,

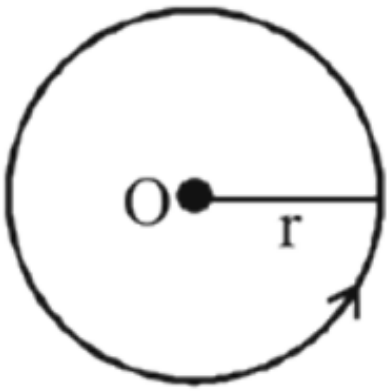
$$\Rightarrow \mathrm{B} \times 2\pi\frac{\mathrm{a}}{2} = \mu_0 \times \frac{1}{\pi \mathrm{a}^2} \times \left(\frac{\mathrm{a}}{2}\right)^2$$
$$= \mu_0 \mathrm{I} \times \frac{1}{4} = \frac{\mu_0 \mathrm{I}}{4}$$
$$\Rightarrow \mathrm{B}_1 = \frac{\mu_0 \mathrm{I}}{4\pi \mathrm{a}}$$
$$\mathrm{B}' \times 2\pi(2\mathrm{a}) = \mu_0 \mathrm{I}$$
$$\frac{\mathrm{B}}{\mathrm{B}'} = \frac{\mu_0 \mathrm{I}}{4\pi \mathrm{a}} \times \frac{4\pi \mathrm{a}}{\mu_0 \mathrm{I}} = 1$$

Q.3 Answer:

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$\frac{\mu_0 \text{ ne}}{2r}$

Solution:



Radius of circular orbit No. of rotations per second

i.e., $\mathrm{T} = \frac{1}{\mathrm{n}}$

Magnetic field at its centre, As we know, current $\mathrm{i} = \frac{\mathrm{e}}{\mathrm{T}} = \frac{\mathrm{e}}{(1/\mathrm{n})} = \mathrm{en} =$ **equivalent current**

Magnetic field at the centre of circular orbit, $\mathrm{B_c} = \frac{\mu_0 \mathrm{i}}{2\mathrm{r}} = \frac{\mu_0 \mathrm{ne}}{2\mathrm{r}}$

Q.4 Answer:

$\mathrm{B_4} > \mathrm{B_1} > \mathrm{B_2} > \mathrm{B_3}$

Solution:

For B_1 **and** B_4 **the contribution due to the different sections add up.**

For B_2 **and** B_3**, the contribution due to the outer sections oppose the contributions due to the inner sections.**

Thus, B_1 **and** B_4 **are greater than** B_2 **and** B_3**.**

For B_4**, there is a section with radius** $< \mathrm{b}$ **and hence, it contributes more than the semicircular section of radius b for** B_1**. Thus,** $B_4 > B_1$

For B_3**, there is section with radius** $> \mathrm{b}$ **and hence it contributes less than semicircular section of radius b for** B_2**. Thus,** $B_3 < B_2$**.**

Hence, $B_4 > B_1 > B_2 > B_3$

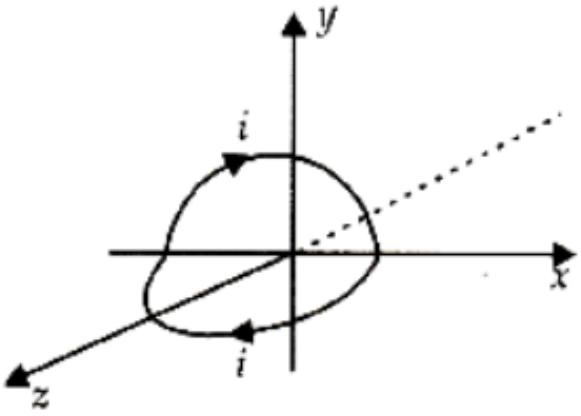
Q.5 Answer:

$\frac{\mu_0 i}{2\sqrt{2}R}$

Solution:

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The loop mentioned in the question must look like one as shown in the figure.



Magnetic field at the centre due to semicircular loop
lying in $x-y$ plane, $B_{xy}=\frac{1}{2}\left(\frac{\mu_0 i}{2R}\right)$ negative z direction.
Similarly field due to loop in $x-z$ plane, $B_{xz}=\frac{1}{2}\left(\frac{\mu_0 i}{2R}\right)$ in negative y direction.
$\therefore$ Magnitude of resultant magnetic field,

$$B=\sqrt{B_{xy}^2+B_{xz}^2}=\sqrt{\left(\frac{\mu_0 i}{4R}\right)^2+\left(\frac{\mu_0 i}{4R}\right)^2}$$
$$=\frac{\mu_0 i}{4R}\sqrt{2}=\frac{\mu_0 i}{2\sqrt{2}R}$$

Q.6 Answer:

$\frac{\mu_0 \mathrm{I}}{2\pi \mathrm{d}}\left(\cos\phi_1+\cos\phi_2\right)$

Solution:

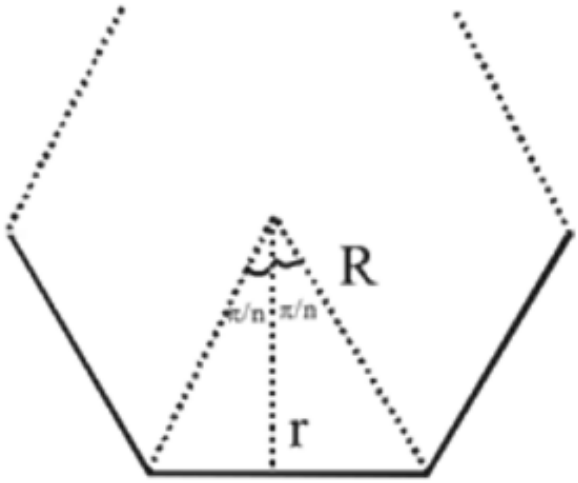
Using B $=\frac{\mu_0 \mathrm{I}}{4\pi \mathrm{d}}\left[\sin\theta_1+\sin\theta_2\right]$
But $\theta_1+\phi_1=90^\circ$ **or** $\theta_1=90^\circ-\phi$,
$\sin\theta_1=\sin\left(90^\circ-\phi_1\right)=\cos\phi_1$
Similarly, $\sin\theta_2=\cos\phi_2$
$B_1=\frac{\mu_0 F}{4\pi d}\left(\cos\phi_1+\cos\phi_2\right)$
$B_{ngt}=2\times\frac{\mu_0\pi}{4\pi d}\left(\cos\phi_1+\cos\phi_2\right)$
$\mathrm{B_{net}}=\frac{\mu_0 \mathrm{I}}{2\pi \mathrm{d}}\left(\cos\phi_1+\cos\phi_2\right)$

Q.7 Answer:

$\frac{\mu_0 \mathrm{I}}{2\pi \mathrm{R}}\left(\tan\frac{\pi}{\mathrm{n}}\right)$

Solution:

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$B = \frac{\mu_0 I}{4\pi r}\left[2\sin\frac{\pi}{n}\right]$

$\cos\frac{\pi}{n} = \frac{r}{R}$

But or $r = R\cos\frac{\pi}{n}$

$\therefore \mathrm{B} = \frac{\mu_0 \mathrm{I}}{4\pi \mathrm{R}\cos\frac{\pi}{\mathrm{n}}}\left[2\sin\frac{\pi}{\mathrm{n}}\right] = \frac{\mu_0 \mathrm{I}}{2\pi \mathrm{R}}\left[\tan\frac{\pi}{\mathrm{n}}\right]$

Q.8 Answer:

$\mu_0 \pi r^3 n \frac{\sigma}{y^3}$

Solution:

Current, $I = \frac{2\pi r\sigma}{\ell/n} = 2\pi r\sigma n$

$B = \frac{\mu_0 I r^2}{2(y^2+r^2)^{3/2}} = \frac{\mu_0 I r^2}{2y^3}$

If $y >> r$.

$B = \frac{\mu_0 r^2}{2y^2}(2\pi r\sigma n) = \frac{\mu_0 \pi r^3 \sigma n}{y^3}$

Q.9 Answer:

$\frac{\mu_0}{2\pi \mathrm{d}}\left(\mathrm{I}_1^2 + \mathrm{I}_2^2\right)^{1/2}$

Solution:

Net magnetic field, $\mathrm{B} = \sqrt{\mathrm{B}_1^2 + \mathrm{B}_2^2}$

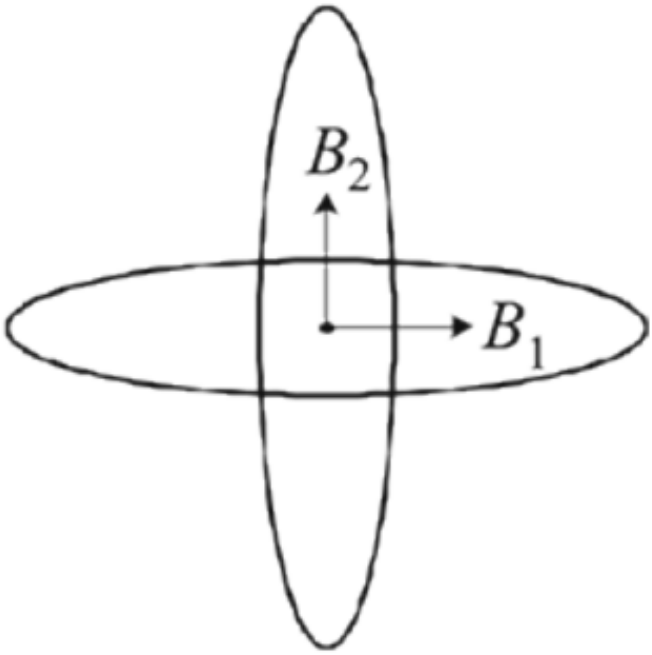
$= \sqrt{\left(\frac{\mu_0 I_1}{2\pi d}\right)^2 + \left(\frac{\mu_0 I_2}{2\pi d}\right)^2} = \frac{\mu_0}{2\pi d}\sqrt{I_1^2 + I_2^2}$

Q.10 Answer:

$\frac{\sqrt{5}\mu_0 I}{2R}$

Solution:

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The magnetic field, due the coil, carrying current I Ampere $B_1 = \frac{\mu_0 I}{2R}$

The magnetic field

due to the coil, carrying

current $2I$ Ampere

$B_2 = \frac{\mu_0(2I)}{2R}$

The resultant B $B_{\text{net}} = \sqrt{B_1^2 + B_2^2 + 2B_1 B_2 \cos\theta}$,$\theta = 90^\circ$

$B_{\text{net}} = \sqrt{B_1^2 + B_2^2} = \frac{\mu_0(2I)}{2R}\sqrt{1+4} = \frac{\sqrt{5}\mu_0 I}{2R}$

Q.11 Answer:

$\frac{\mu_0}{\pi} \cdot \frac{\mathrm{i}}{\ell}\sqrt{3}$

Solution:

$$B \text{ due to AB} = \frac{\mu_0}{4\pi}\frac{i}{d}[\sqrt{3}]$$

$$= \frac{\mu_0}{4\pi}\frac{i}{\ell/2}[\sqrt{3}] = \frac{\mu_0 i}{4\pi\ell}2\sqrt{3}$$

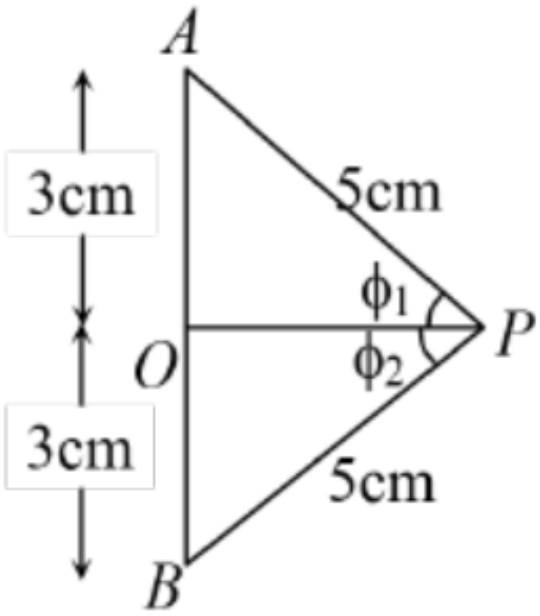
$$\text{Net magnetic field } = \frac{\mu_0}{2\pi}\frac{i}{\ell}\sqrt{3} \times 2 = \frac{\mu_0}{\pi}\frac{i}{\ell}\sqrt{3}$$

Q.12 Answer:

0.15 gauss

Solution:

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Now $B = \frac{\mu_0 I}{4\pi r}(\sin\phi_1 + \sin\phi_2)$

Here $\mathbf{r = OP}$

Now, $AO = OB = \frac{6}{2}$ cm $= 3$ cm **and** $PB = PA = 5$ cm

$\therefore \quad OP = \sqrt{(PB)^2 - (OB)^2} = 4$ cm

$\therefore \quad r = 4$ cm $= 4 \times 10^{-2}$ m

$\sin\phi_1 = \frac{3}{5}$ **and** $\sin\phi_2 = \frac{3}{5}$

$\therefore$ **(c)**

Q.13 Answer:

$\frac{5\mu_0 I\theta}{24\pi r}$

Solution:

Since magnetic field at the centre of an arc is equal to $B = \frac{\mu_0 I}{4\pi r}\theta$

Hence, net $\mathbf{B} = \frac{\mu_0 I}{4\pi}\left(\frac{1}{r} - \frac{1}{2r} + \frac{1}{3r}\right)\theta = \frac{5\mu_0 I\theta}{24\pi r}$

$\therefore$ **(n)**

Q.14 Answer:

R

Solution:

$\frac{\mu_0 iR^2}{2(R^2+x^2)^{3/2}} = \frac{1}{\sqrt{8}}\frac{\mu_0 i}{2R}, \quad \left(R^2+x^2\right)^{3/2} = R^3\sqrt{8}$

$\left(R^2+x^2\right)^3 = 8R^6, \quad R^2+x^2 = 2R^2, \quad x = \pm R$

(d)

Q.15 Answer:

5×10^{-4}

Solution:

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$B_{\text{due to straight wire}} = \frac{\mu_0 i}{2\pi r} = \frac{2\times10^{-7}\times30}{2\times10^{-2}} = 3\times10^{-4}$ **Tesla**

$B_{\text{external}} = 4\times10^{-4}$ **Tesla**

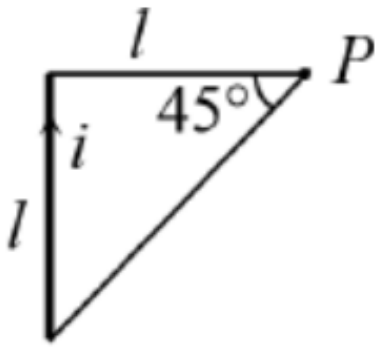
$B_{\text{net}} = \sqrt{3^2+4^2}\times10^{-4} = 5\times10^{-4}$ **Tesla**

$\therefore$ **(c)**

Q.16 Answer:

$\frac{\sqrt{2}\mu_0 i}{8\pi l}$

Solution:



$B = \frac{\mu_0}{4\pi}\cdot\frac{i}{l}(\sin\phi_1 + \sin\phi_2)$

Here, $\phi_1 = 0^\circ, \phi_2 = 45^\circ \Rightarrow B = \frac{\sqrt{2}\mu_0 i}{8\pi l}$

$\therefore$ **(c)**

Q.17 Answer:

$\frac{3}{2}$

Solution:

$$B_1 = -\frac{\mu_0 i_1}{2\pi r/2} + \frac{\mu_0 i_2}{2\pi r/2} = \frac{\mu_0}{\pi r}(i_2 - i_1)$$

$$= 6\times10^{-6}\ \mathrm{T}$$

When the current is reversed in I_2**,**

$\mathrm{B}_2 = -\frac{\mu_0(i_1+i_2)}{2\pi r/2} = -\frac{\mu_0(i_1+i_2)}{\pi r} = 3\times10^{-5}\ \mathrm{T}$

Dividing (ii) by (i) we get

$\frac{-(i_1+i_2)}{i_2-i_1} = \frac{30}{6} = 5$

$-(i_1+i_2) = 5i_2 - 5i_1 \Rightarrow 6i_2 = 4i_1$

$\frac{i_1}{i_2} = \frac{3}{2}$

Q.18 Answer:

$\overrightarrow{\mathrm{B}} = -\frac{\mu_0}{4\pi}\frac{\mathrm{I}}{\mathrm{R}}(\hat{\pi}\hat{\mathrm{i}} + 2\hat{\mathrm{k}})$

Solution:

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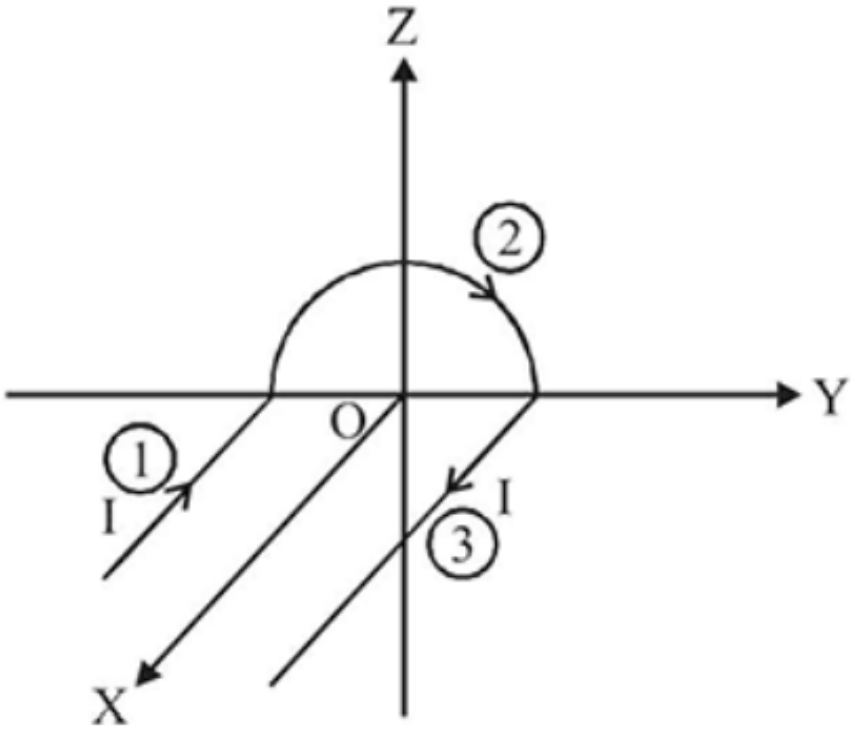
Magnetic field due to segment ' 1 '

$\overrightarrow{\mathrm{B}_1}=\frac{\mu_0 \mathrm{I}}{4 \pi \mathrm{R}}\left[\sin 90^{\circ}+\sin 0^{\circ}\right](-\hat{\mathrm{k}})$

$=\frac{-\mu_0 \mathrm{I}}{4 \pi \mathrm{R}}(\hat{\mathrm{k}})=\overrightarrow{\mathrm{B}}_3$

Magnetic field due to segment 2

$\mathrm{B}_2=\frac{\mu_0 \mathrm{I}}{4 \mathrm{R}}(-\hat{\mathrm{i}})=\frac{-\mu_0 \mathrm{I}}{4 \pi \mathrm{R}}(\pi \hat{\mathrm{i}})$



$\therefore \overrightarrow{\mathrm{B}}$ **at centre**

$\overrightarrow{\mathrm{B}}=\overrightarrow{\mathrm{B}}_1+\overrightarrow{\mathrm{B}}_2+\overrightarrow{\mathrm{B}}_3=\frac{-\mu_0 \mathrm{I}}{4 \pi \mathrm{R}}(\pi \hat{\mathrm{i}}+2 \hat{\mathrm{k}})$

Q.19 Answer:

$1.5 \times 10^{-2}\ \mathrm{N}-\mathrm{m}$

Solution:

Torque on the solenoid is given by $\tau = MB\sin\theta$

where θ **is the angle between the magnetic field and the axis of solenoid.** $M = \mathrm{niA}$

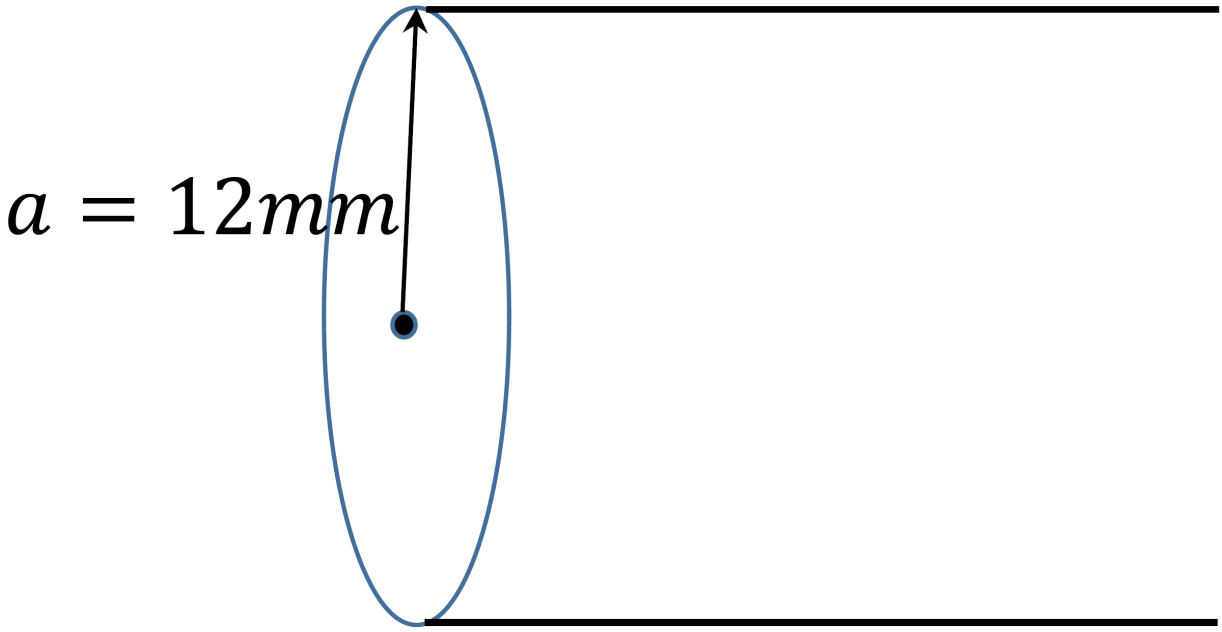
$$\therefore \tau = \mathrm{niA}\, B \sin 30^{\circ}$$
$$= 2000 \times 2 \times 1.5 \times 10^{-4} \times 5 \times 10^{-2} \times \frac{1}{2}$$
$$= 1.5 \times 10^{-2} N - m$$

Q.20 Answer:

10

Solution:

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By Ampere's law;

$\oint B\,.\,dl = \mu_0\,.\,I \qquad \ldots\left(i\right)$

Here, the line integral of the length will be

$\oint dl = 2\pi r$

$\Rightarrow \oint B\,.\,dl = B\,.\,2\pi\ \frac{a}{2}$

Now, the total current in the wire will be the integral of current density over the whole area.

$I = \int J\,.\,da$

Putting the value of I and $\oint B\,.\,dl$ in equation i.

$\Rightarrow \quad B \times \frac{2\pi a}{2} = \mu_0 \int_0^{\frac{a}{2}} \frac{J_0 r}{a} \times 2\pi r dr \quad \because da = 2\pi r dr$

The integration is to be taken from $0 \quad to \quad \frac{a}{2}$ **for the current density.**

$\pi a B = \frac{2\pi\mu_0 J_0}{a} \times \frac{\frac{a}{2}^3}{3}$

$B = \frac{\mu_0 J_0 a}{12}, \mu_0 = 4\pi \times 10^{-7}$

$\Rightarrow B = \frac{4\pi \times 10^{-7} \times 10^5 \,/\, 4\pi \times 0.012}{12}$

$\therefore B = 10 \quad \mu T$

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